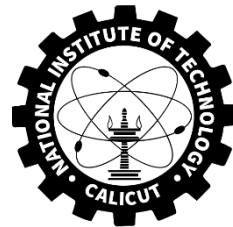


Quality, Reliability & Maintenance ME3327E



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Quality Systems and Standards



- The three standards of ISO 9000 are:
 - ISO 9000:2000 Quality Management System—Fundamentals and Vocabulary
 - ISO 9001:2000 Quality Management System—Requirements
 - ISO 9004:2000 Quality Management System—Guidelines for Performance Improvement
- ISO 9000 is also an American National Standards Institute and an ASQ standard.
- The ISO 9001:2000 standard has eight clauses:
 1. Scope
 2. Normative References
 3. Definitions
 4. Quality Management Systems
 5. Management Responsibility
 6. Resource Management
 7. Product (or Service) Realization
 8. Measurement, Analysis, and Improvement.

The Link Between Quality and Productivity

- Effective quality improvement can be instrumental in increasing productivity and reducing cost.
- When technological advances occur rapidly and when the new technologies are used quickly to exploit competitive advantages, the problems of designing and manufacturing products of superior quality are greatly complicated.
- Often, too little attention is paid to achieving all dimensions of an optimal process: economy, efficiency, productivity, and quality.
- Effective quality improvement can be instrumental in increasing productivity and reducing cost.
- An engineering study of this process reveals that excessive process variability is responsible for the extremely high fallout.
- Note that the installation of statistical process control and the reduction of variability that follows result in reduction in manufacturing costs.

Product and Service Cost

Direct and Indirect Cost

- *Direct costs*, such as direct labor and materials, are a function of the number of units of the manufactured product or the number of customers serviced.
- *Indirect cost*, do not change with each unit produced or each customer served, such as machine setup for the same product, depreciation of building, property taxes, and so on.
- Indirect costs should be distributed to products or services based on cause-and-effect relations or actual use.
- Traditional accounting methods can lead to misleading product/service costs where indirect costs are allocated based on direct labor or direct material.
- A better estimate of product costing is arrived at by using activities that measure the degree of use of the particular resource. This is known as **activity-based costing**.

Activity Based Costing

- *Activities* are tasks performed by a specialized group or department, say the purchasing unit in an organization, also known as an *activity* or *cost center*.
- The types of transactions that generate costs are identified as *cost drivers*.
- Cost drivers are typically categorized into four groups.
 - Unit-level costs comprise those activities that are associated with each product/service unit.
 - Batch-level costs are based on activities that are performed once for each lot or batch of products and are not influenced by the number of units in the batch.
 - Product/service-level cost, which is based on the type of product/service.
 - Production/service-sustaining cost level, which incorporates activities that use all other resources necessary to maintain operations.

Exercise

Two types of microchips (A and B) are being manufactured, with micro chip B being slightly more complex. There are two sizes of microchip A, A1 and A2. For each type and each size, microchips are manufactured in batch sizes of 100,000. Table 1 shows the production volume and direct costs/batch, while Table 2 displays the overhead costs for the past year. Using the traditional costing method (unit-based costing) of allocating overhead rate based on direct labor costs, calculate the cost per batch of each type of microchip. Calculate the cost per batch using the activity-based costing method and compare with the figures calculated previously.

TABLE 1 Production Volm & Direct Costs per Batch of Microchips

	Microchip A		Microchip B
	A1	A2	
No. of batches produced	500	800	1200
Cost/Batch (in INR Cr.)			
Direct Labor	0.05	0.06	0.08
Direct Material	0.25	0.32	0.38
Processing	0.15	0.2	0.3

TABLE 2 Overhead Cost of Plant

Category	Cost (INR Cr.)
Setup and Testing	22
Product Lining cost	
Microchip A	55
Microchip B	93
Other plant cost	45
Total overhead cost	215

Quality Cost

- Financial controls are an important part of business management.
- These financial controls involve a comparison of actual and budgeted costs, along with analysis and action on the differences between actual and budget.
- Many organizations now formally evaluate the cost associated with quality.
- There are several reasons why the cost of quality should be explicitly considered in an organization. These include the following:
 - The increase in the cost of quality because of the increase in the complexity of manufactured products associated with advances in technology
 - Increasing awareness of life-cycle costs, including maintenance, spare parts, and the cost of field failures
 - Quality engineers and managers can most effectively communicate quality issues in a way that management understands.

Quality Cost

Prevention Costs

- Quality planning and engineering
- New products review
- Product/process design
- Process control
- Burn-in
- Training
- Quality data acquisition and analysis

Appraisal Costs

- Inspection and test of incoming material
- Product inspection and test
- Materials and services consumed
- Maintaining accuracy of test equipment

Internal Failure Costs

- Scrap
- Rework
- Retest
- Failure analysis
- Downtime
- Yield losses
- Downgrading (off-specing)

External Failure Costs

- Complaint adjustment
- Returned product/material
- Warranty charges
- Liability costs
- Indirect costs

Quality Costs Data Requirement

**Senior
Management**

- Selling Price
- Marketing Cost
- General Administrative Cost

**Middle
Management**

- Indirect Labor
- Indirect Material
- Fixed Costs

Line Management

- Direct Labor
- Direct Material

Legal Aspects of Quality

- Consumerism and product liability are important reasons why quality assurance is an important business strategy.
- Consumerism is in part due to the seemingly large number of failures in the field of consumer products and the perception that service quality is declining.
- Product liability is a major social, market, and economic force.
- The legal obligation of manufacturers and sellers to compensate for injury or damage caused by defective products is not a recent phenomenon.
- The concept of product liability has been in existence for many years, but its emphasis has changed recently.

Implementing Quality Improvement

- Strategic management of quality in an organization must involve all three components: quality planning, quality assurance, and quality control and improvement.
- Furthermore, all of the individuals in the organization must have an understanding of the basic tools of quality improvement.
- It is increasingly important that everyone in an organization, from top management to operating personnel, have an awareness of basic statistical methods and of how these methods are useful in manufacturing, engineering design, and development and in the general business environment.

Assignment

A company manufacturing a chemical compound has identified the following cost components (per kilogram): direct labor, ₹ 200; material, ₹ 500; energy, ₹ 50; overhead, 30% of direct labor and material. The process is operating at a yield rate of 96%. A quality improvement team, through analysis, has been able to increase yield to 97%. Find the cost per kilogram of product before and after this improvement. On further study, the team analyzed the process and was able to eliminate several non-value-added steps. Direct labor and material costs were reduced by 25%. Calculate the cost per kilogram of product after these process changes are made. By what percentage have total costs been reduced? What is the percentage increase in capacity?

Assignment



In a doctor's office with an office administrator and a nurse, the average waiting time for patients prior to being seen by the physician is 15 minutes. The average examination time by the physician, per patient, is 10 minutes. Currently, the office administrator schedules patients every 15 minutes for a 6-hour period. When salary and benefits are converted to an hourly rate, for a nurse it is ₹ 40/hour, while for an office administrator it is ₹ 20/hour. The practice, which treats Medicare/Medicaid patients, receives an average reimbursement of ₹ 400 per patient visit. Excluding the physician's salary and other overhead costs associated with equipment, discuss the current profitability of operations of the office. Assume that the office staff work 8 hours/day. Effective process improvement measures were taken by incorporating electronic medical records and check-in via the Internet. Such steps reduced average waiting time per patient to 10 minutes. If the office administrator now schedules patients every 10 minutes, what is the profitability measure? Assume that the physician will see all scheduled patients for the day.